$$k = \left(\frac{20}{20}\right) = 1$$

Rate constant for a reaction = 1

ii) To calculate the time for 80% of completion

k = 1, a = 100, x = 80%, t = ?

Therefore,

$$t = \left(\frac{x}{k}\right) = \left(\frac{80}{1}\right) = 80\,\text{min}$$

27 Solution:

Here, we are given that

E_a = 22.5 kcal mol^{-1} = 22500 cal mol^{-1}

T = 40°C = 40 + 273 = 313 K

k = 1.8 × 10^{-5} sec^{-1}

Substituting the values in the equation

$$\log A = \log k + \left(\frac{E_a}{2.303RT}\right)$$

$$\log A = \log\left(1.8\times10^{-5}\right) + \left(\frac{22500}{2.303\times1.987\times313}\right)$$

$$\log A = \log\left(1.8\right) - 5 + \left(15.7089\right)$$

$$\log A = 10.9642$$

$$A = \text{antilog}\left(10.9642\right)$$

$$A = 9.208\times10^{10}\,\text{collisions}\ s^{-1}$$

28.**Solution:**

For a first order reaction

$$k = \frac{2.303}{t}\log\frac{a}{(a-x)}$$

$$k = \frac{2.303}{t}\log\frac{V_\infty}{V_\infty - V_t}$$

In the present case, V_∞ = 58.3 ml.

The value of *k* at different time can be calculated as follows:

t (*min*)	V_t	$V_\infty - V_t$	$\frac{2.303}{t}\log\frac{V_\circ}{V_\circ - V_t}$
6	19.3	58.3-19.3 = 39.0	$k = \frac{58.3}{\ldots}\log\frac{58.3}{25.7} = 0.0670\ \text{min}^{-1}$
12	32.6	58.3-32.6 = 25.7	$k = \frac{58.3}{\ldots}\log\frac{58.3}{25.7} = 0.0683\ \text{min}^{-1}$
18	41.3	58.3-41.3 = 17.0	$k = \frac{58.3}{11.8}\log\frac{58.3}{11.8} = 0.0685\ \text{min}^{-1}$
24	46.5	58.3-46.5 = 11.8	$K = \frac{58.3}{11.8}\log\frac{58.3}{11.8} = 0.0666\ \text{min}^{-1}$

Since the value of *k* comes out to be nearly constant, the given reaction is of the first order. The mean value of *k* = 0.0676 min^{-1}

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